exercise_3.py

```python
## Course 1: Natural Language Processing
## Module 1: Intro to NLP in Python
## Coding Exercise 3
from nltk.tokenize import word_tokenize
from nltk import pos_tag
from nltk.corpus import stopwords

def find_pronouns(sentences):
  pronouns = []
  for sentence in sentences:
    words = word_tokenize(sentence)
    words = [words for words in word_tokenize(sentence) if words not in set(stopwords.
    for word, pos in pos_tag(words):
      #print(word, pos)
      if pos == 'PRP':
        pronouns.append(word)
  return pronouns

sentences = ['They are good at playing football.', \
             'He has many backpacks.', \
             'She has lots of pens in her bag.', \
             'I forgot my ID.', \
             'The dog stared at himself in the reflection.', \
             'She told me it had to be done by end of day.']
print(find_pronouns(sentences))
```

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Guide

Collapse

Coding Exercise 3

Problem

Create the function `find_pronouns` that takes a list of strings and returns a list of all the personal pronouns.

Expected Output

After creating the `find_pronouns` function, pass it the following variable and print the return value.

```python
sentences = ['They are good at playing football.', 'He has many backpacks.', 'She h
```

Click the `TRY IT` button to run the file and check your work. The expected output is shown below.

TRY IT

```
['They', 'He', 'She', 'I', 'She']
```

```
['They', 'He', 'She', 'I', 'She']
```

Submit your work for evaluation.

Here is one possible solution:

```python
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk import pos_tag

sentences = ['They are good at playing football.', 'He has many backpack

def find_pronouns(data):
  pronouns = []
  for sentence in data:
    tokens_no_sw = [token for token in word_tokenize(sentence) if token
    tags = pos_tag(tokens_no_sw)
    for (token, tag) in tags:
      if tag == 'PRP': pronouns.append(token)
  return(pronouns)

print(find_pronouns(sentences))
```

- Import `stopwords`, `word_tokenize`, `sent_tokenize`, and `pos_tag` from the `nltk` module
- Create the `find_pronouns` function
- Create the empty list `pronouns`
- Iterate over the list of strings passed to the function
- Create a list of all of the words in the string that are not a stop word
- Tag each word with their part of speech
- Iterate over the list of tagged words
- If the word is a pronoun, append it to the list `pronouns`
- Return `pronouns`

Check It!

LAST RUN on 12/27/2023, 3:28:58 AM

```
['They', 'He', 'She', 'I', 'She']
test_1 (test_exercise_3.test_ex3) ...
ok
test_2 (test_exercise_3.test_ex3) ...
ok
test_3 (test_exercise_3.test_ex3) ...
ok

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```

Next